# Analysis Guiding Questions

When using the guiding questions below, consider that you are taking the role of a XAI system *designer* and the questions address characteristics of the *end-users* for whom you are designing and the *context of use* in which the system will be used.

**Personas**

| Impact Factor(s) | Guiding Question | Situation Illustration |
|---|---|---|
| Decision-maker's domain expertise | 1. What have I learned about your expertise and knowledge in the decision-making domain? Are you an expert or novice? | Experts have more knowledge in the decision-making domain and tend to question more the AI recommendation. Novices tend more to follow AI advice. |
| Task familiarity | 2. What is your familiarity with the task I am designing for? | When a decision-maker has high familiarity with a decision-making task, he/she will know better what to expect from the AI recommendation, different from someone with low familiarity. |
| AI literacy | 3. What have I learned about your knowledge on artificial intelligence systems? Are you a lay user or do you understand fundamentals on how these systems work? | AI literacy refers to the user's understanding of AI limitations and mechanisms. A user with knowledge on AI will be able to expect the potential behavior of the system and question its output. A lay user may follow blindly the AI advice. |
| AI familiarity | 4. How familiar are you with the use of artificial intelligence systems? Have you interacted with them in a regular basis or is it your first experience? | AI familiarity refers to the user's prior experience interacting with AI systems. A user without experience interacting with AI systems will be more cautious to accept its advice, different from a user familiar with these systems. |
| Openness to a new experience | 5. How open are you to try a new experience? And to interact with a new system? Enthusiast, open-minded, or skeptic? | When a user is enthusiast or has an open mind to interact with a new system, he/she will be more open to accept an AI advice, different from skeptical users. |

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| Detect and correct AI's mistakes | 6. What have I learned about your ability to examine the correctness of a recommendation? | The decision-maker's characteristics will frame his/her ability to detect AI mistakes. Someone without this ability can follow blindly AI advice even when it is wrong. |
| Interest in the task | 7. How interested are you in the decision-making task? Are you cautious about following advice when making a decision, or do you not show close attention to it? | A decision-maker who has interest in the current task will be more cautious in following AI advice, different from someone without interest, who won't show close attention. |
| Decision-maker's autonomy | 8. How motivated are you to keep your interests and values when making a decision? | A decision-maker with high autonomy will ensure his/her own opinion when making a decision, different from someone with low autonomy who tends to follows more the AI advice. |
| Algorithm aversion | 9. How averse are you to interact with an artificial intelligence agent? Do you discount a recommendation because it was made by a computing system? Do you appreciate it when you have a positive first impression of the system? | A user with aversion to use computing systems will reject more an AI advice, different from someone without this aversion. |
| Decision-maker's self-confidence | 10. What have I learned about your own confidence for making a decision in the context I am designing for? Do you think to be able to achieve better decisions by yourself instead of using the artificial intelligence recommendations? | A decision-maker with high self-confidence will pay close attention to his/her own decision compared to the AI advice, different from someone with low self-confidence. |
| Decision-maker's cognitive load | 11. What have I learned about your ability to handle excessive information? Do you tend to skip information or analyze each one in detail when making a decision? | A decision-maker who tends to skip information will accept more the AI advice, different from someone who analyzes information in detail and based on that can identify AI mistakes. |

## Scenarios of Use

| Impact Factor(s) | Guiding Question | Situation Illustration |
|---|---|---|
| Field of application | 12. How is the field of application I am designing for characterized? High-risk or low-risk context? | Wrong decisions made in high-risk context, such as medicine, can have greater consequences to the individuals involved, different from low-risk context, such as leisure. |
| Degree of vulnerability | 13. How vulnerable and uncertain are the situations belonging to this context? | In certain contexts, such as medicine and finance, the decisions made can generate vulnerable situations to the individuals involved. Not all contexts have vulnerability. |
| Time pressure for decision-making | 14. How urgent are decisions in this context? Why? | When decisions have to be made under a high time pressure, the decision-maker can accept more the AI advice compared with situations in which there is low or no time pressure. |
| Domain complexity | 15. How complex is the decision-making domain according to what I have learned about your [user] expertise? | A highly complex domain can lead the decision-maker to rely more on AI advice due to its ability to analyze several kinds of information. |
| Task complexity | 16. What is the decision-making task I am design for? How difficult will it be for you to make a decision in this task? | A high difficulty task can also lead novice decision-makers to follow blindly the AI advice, compared to low difficulty tasks and when the decision-maker is expert. |